Supplementary Material: Tables and Proofs

1. Tables

To provide a comprehensive validation of the proposed framework, five tables are included as supporting materials to illustrate different aspects of the model. Table 1 illustrates the conversion of fourth-order clauses into lower-order clauses using the Tseitin transformation. Table 2 presents the clause satisfaction analysis of the QMAX2SAT formulation, where the relationship between neuron states, clause evaluations, and the corresponding energy values is explicitly quantified.

Furthermore, Table 3 provides a detailed illustration of the structural changes in $P_{\text{QMAX2,3SAT}}$ across different transformation stages, including redundancy elimination and Tseitin transformation. Based on this, Table 4 evaluates the impact of QMAX2SAT by comparing the ratios of maximum satisfiable clauses under different neuron states. Finally, Table 5 reports the Jaccard similarity across various transformation stages, offering a quantitative measure of structural consistency.

Together, these tables collectively support the effectiveness and robustness of the proposed approach from logical transformation, clause satisfaction, structural evolution, and similarity perspectives.

Table 1: Truth table verifying the Tseitin transformation of fourth-order clauses into lower-order clauses

$l_i^d \vee l_i^e \vee l_i^f \vee l_i^g$	$l_i^d \vee l_i^e \vee \neg p$	$p \vee l_i^f \vee l_i^g$	$\neg l_i^d \vee p$	$\neg l_i^e \vee p$	$(l_i^d \vee l_i^e \vee \neg p) \wedge (p \vee l_i^f \vee l_i^g) \wedge (\neg l_i^d \vee p) \wedge (\neg l_i^e \vee p)$
1	$l_i^d \vee l_i^e = p = 1$	$l_i^f \vee l_i^g = 1, p = 1$	1	1	1
1	$l_i^d \vee l_i^e = p = 1$	$l_i^f \vee l_i^g = 0, p = 1$	1	1	1
1	$l_i^d \vee l_i^e = p = 0$	$l_i^f \vee l_i^g = 1, p = 0$	1	1	1
0	$l_i^d \vee l_i^e = p = 0$	$l_i^f \vee l_i^g = 0, p = 0$	1	1	0

2. The proof of the theorem

2.1. Proof of Theorem 1

Theorem 1. *For an n-dimensional dataset ($n \geq 2$), where each dimension is characterized by two mutually exclusive attributes, there exists a structure-preserving one-to-one mapping that embeds each information element into a literal of a substructure in $P_{QMAX2,3SAT}$.*

Table 2: Clause satisfaction analysis of the $P_{\text{QMAX2SAT}} = (\neg l_i^p \vee \neg l_i^q) \wedge (\neg l_i^p \vee l_i^q) \wedge (l_i^p \vee \neg l_i^q) \wedge (l_i^p \vee \neg l_i^r) \wedge (l_i^q \vee l_i^r)$

$S_{l_i^p}$	$S_{l_i^q}$	$S_{l_i^r}$	$\neg l_i^p \vee \neg l_i^q$	$E_{\neg l_i^p \vee \neg l_i^q}$	$\neg l_i^p \vee l_i^q$	$E_{\neg l_i^p \vee l_i^q}$	$l_i^p \vee \neg l_i^q$	$E_{l_i^p \vee \neg l_i^q}$	$l_i^p \vee \neg l_i^r$	$E_{l_i^p \vee \neg l_i^r}$	$l_i^q \vee l_i^r$	$E_{l_i^q \vee l_i^r}$	$E_{P_{\text{QMAX2SAT}}}$	$N_{P_{\text{QMAX2SAT}}}$
1	1	1	-1	1	1	0	1	0	1	0	1	0	1	4
1	1	-1	-1	1	1	0	1	0	1	0	1	0	1	4
1	-1	1	1	0	-1	1	1	0	1	0	1	0	1	4
1	-1	-1	1	0	-1	1	1	0	1	0	-1	1	2	3
-1	1	1	1	0	1	0	-1	1	-1	1	1	0	2	3
-1	1	-1	1	0	1	0	-1	1	1	0	1	0	1	4
-1	-1	1	1	0	1	0	1	0	-1	1	1	0	1	4
-1	-1	-1	1	0	1	0	1	0	1	0	-1	1	1	4

Note: $E_{\neg l_i^p \vee \neg l_i^q} = \frac{1}{2}\left(1 + S_{l_i^p}\right) \cdot \frac{1}{2}\left(1 - S_{l_i^q}\right)$ is the cost function of $\neg l_i^p \vee \neg l_i^q$ defined in this paper.

Table 3: Illustration of the conversion process for the example $P^3_{\text{QMAX2,3SAT}}$ across different transformation stages.

$P_{\text{QMAX2,3SAT}}$	Initial formulation				(1) Redundancy elimination				(2) Tseitin transformation				After (1) and (2)			
	N_c^o	N_c^c	N_c^{sc}	N_c^l	N_c^o	N_c^c	N_c^{sc}	N_c^l	N_c^o	N_c^c	N_c^{sc}	N_c^l	N_c^o	N_c^c	N_c^{sc}	N_c^l
$C_{ik}^{(\text{QMAX2SAT})}$	1	5	4	3	1	5	4	3	1	5	4	3	1	5	4	3
$C_i^{(2)}$	1	1	1	2	1	1	1	2	1	1	1	2	1	1	1	2
$C_i^{(3)}$	1	1	1	3	1	1	1	3	1	1	1	3	1	1	1	3
$C_{i1}^{(3)} \wedge C_{i1}^{(3)}$	1	2	2	3	1	1	1	2	1	2	2	3	1	1	1	2
$C_i^{(4)}$	1	1	1	4	1	1	1	4	1	4	4	5	1	4	4	5
Overall	5	10	9	15	5	9	8	14	5	13	12	16	5	12	11	15

Note: $P^{(3)}_{\text{QMAX2,3SAT}} = (\neg l_1^p \vee \neg l_1^q) \wedge (\neg l_1^p \vee l_1^q) \wedge (l_1^p \vee \neg l_1^q) \wedge (l_1^p \vee \neg l_1^r) \wedge (\neg l_1^q \vee l_1^r) \wedge (l_1^s \vee l_1^t) \wedge (l_1^x \vee l_1^y \vee l_1^z) \wedge (l_1^a \vee l_1^b \vee l_1^c) \wedge (l_1^a \vee l_1^b \vee \neg l_1^c) \wedge (l_1^d \vee l_1^e \vee l_1^f \vee l_1^g)$

$\Longleftrightarrow (\neg l_1^p \vee \neg l_1^q) \wedge (\neg l_1^p \vee l_1^q) \wedge (l_1^p \vee \neg l_1^q) \wedge (l_1^p \vee \neg l_1^r) \wedge (\neg l_1^q \vee l_1^r) \wedge (l_1^s \vee l_1^t) \wedge (l_1^x \vee l_1^y \vee l_1^z) \wedge (l_1^a \vee l_1^b) \wedge (l_1^d \vee l_1^e \vee \neg l_1^{p^{d\vee e}}) \wedge (l_1^{p^{d\vee e}} \vee l_1^f \vee l_1^g) \wedge (\neg l_1^d \vee l_1^{p^{d\vee e}}) \wedge (\neg l_1^e \vee l_1^{p^{d\vee e}})$.

Table 4: The impact of QMAX2SAT: The ratio of the maximum satisfiable clauses in different situations of the example in Table 3.

Neuron states of $P_{\text{QMAX2,3SAT}}$	Initial formulation			(1) Redundancy elimination			(2) Tseitin transformation			After (1) and (2)		
	R_s^o	R_s^c	R_s^{sc}	R_s^o	R_s^c	R_s^{sc}	R_s^o	R_s^c	R_s^{sc}	R_s^o	R_s^c	R_s^{sc}
(1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1)	1	$\frac{9}{10}$	1	1	$\frac{8}{9}$	1	1	$\frac{12}{13}$	1	1	$\frac{11}{12}$	1
(1,-1,-1,1,1,1,1,1,1,1,1,1,1,1,1,1)	$\frac{4}{5}$	$\frac{8}{10}$	$\frac{8}{9}$	$\frac{4}{5}$	$\frac{7}{9}$	$\frac{7}{8}$	$\frac{4}{5}$	$\frac{11}{13}$	$\frac{11}{12}$	$\frac{4}{5}$	$\frac{10}{12}$	$\frac{10}{11}$
(1,1,1,1,1,1,1,1,-1,-1,-1,1,1,1,1)	$\frac{4}{5}$	$\frac{7}{10}$	$\frac{7}{9}$	$\frac{4}{5}$	$\frac{7}{9}$	$\frac{7}{8}$	$\frac{4}{5}$	$\frac{10}{13}$	$\frac{10}{12}$	$\frac{4}{5}$	$\frac{10}{12}$	$\frac{10}{11}$
(1,1,1,1,1,1,1,1,1,1,1,-1,-1,-1,-1)	$\frac{4}{5}$	$\frac{8}{10}$	$\frac{8}{9}$	$\frac{4}{5}$	$\frac{7}{9}$	$\frac{7}{8}$	$\frac{4}{5}$	$\frac{11}{13}$	$\frac{11}{12}$	$\frac{4}{5}$	$\frac{10}{12}$	$\frac{10}{11}$
(1,1,1,-1,-1,1,1,1,1,1,1,1,1,1,1,1)	$\frac{4}{5}$	$\frac{8}{10}$	$\frac{8}{9}$	$\frac{4}{5}$	$\frac{7}{9}$	$\frac{7}{8}$	$\frac{4}{5}$	$\frac{11}{13}$	$\frac{11}{12}$	$\frac{4}{5}$	$\frac{10}{12}$	$\frac{10}{11}$
(1,1,1,1,1,-1,-1,-1,1,1,1,1,1,1,1,1)	$\frac{4}{5}$	$\frac{8}{10}$	$\frac{8}{9}$	$\frac{4}{5}$	$\frac{7}{9}$	$\frac{7}{8}$	$\frac{4}{5}$	$\frac{11}{13}$	$\frac{11}{12}$	$\frac{4}{5}$	$\frac{10}{12}$	$\frac{10}{11}$

Table 5: Jaccard similarity of the example in Table 3

Neuron states of $P_{\text{QMAX2,3SAT}}$	Initial formulation	(1) Redundancy elimination	(2) Tseitin transformation	After (1) and (2)
(1,1,1,1,1,1,1,1,1,1,1,1,1,1,1,1)	$\frac{18}{25}$	$\frac{15}{21}$	$\frac{21}{31}$	$\frac{18}{27}$

Proof. QMAX2,3SAT is a hybrid logical rule that integrates second-, third-, and fourth-order clauses together with a QMAX2SAT component consisting of five second-order clauses and three literals, all formulated in conjunctive normal form (CNF). To establish the mapping between the dataset and the QMAX2,3SAT structure, we consider the following two cases:

Case 1: When n is even, each dimension can be paired and represented by a set of second-order clauses. Hence, the n-dimensional dataset can be expressed as an even multiple of binary relations:

$$n = 2 + 2 + \cdots + 2 = 2 \times \frac{n}{2}. \tag{1}$$

Case 2: When n is odd and $n \geq 3$, one third-order clause can be introduced, and the remaining $(n-3)$ dimensions form an even structure. Therefore,

$$n = 3 + (n-3), \tag{2}$$

where $(n-3)$ can be fully represented by combinations of second-order clauses.

From the above cases, it follows that for any $n \geq 2$, there exists at least one QMAX2,3SAT substructure capable of encoding an n-dimensional dataset, in which each dimension variable establishes a one-to-one correspondence with a literal in the logical rule. Given that each dimension variable admits two possible states, the bipolar representation $l_i \in \{-1, 1\}$ adopted in QMAX2,3SAT provides a natural and consistent encoding.

Furthermore, the application of the Tseitin transformation preserves satisfiability equivalence during structural reduction, ensuring that the logical properties of the original formulation remain intact. It is noted that the above construction demonstrates the existence of a valid mapping; in practice, multiple distinct QMAX2,3SAT substructures may satisfy the same dataset-to-logic correspondence.

This completes the proof. □

2.2. Proof of Theorem 2

Theorem 2. *Under random initialization of neuron states, the probability that the induced neural configuration achieves the* maximum clause satisfaction *of the $P_{\text{QMAX2,3SAT}}$ is characterized by*

$$\Pr(\theta(P_{\text{QMAX2,3SAT}}) = \theta_{\max}) = \left(\frac{3}{4}\right)^{v+w+m} \left(\frac{7}{8}\right)^{n} \left(\frac{15}{16}\right)^{u}, \tag{3}$$

where the exponents v, w, m, n, and u correspond to the respective numbers of integrated logical components arising from $P_{\text{QMAX2,3SAT}}$ clause fusion.

Proof. Each neuron in the network takes bipolar values $S_i \in \{1, -1\}$, which are directly interpreted as *true* and *false*, respectively. Consider a clause $C_i^{(2)} = l_i^s \vee l_i^t$ in $P_{\text{QMAX2,3SAT}}$. The probability that l_i^s is *false* equals $\frac{1}{2}$, and the same holds for l_i^t. Since $C_i^{(2)}$ is the disjunction of l_i^s and l_i^t, we obtain

$$\begin{aligned} \Pr\left(C_i^{(2)} = -1\right) &= \Pr\left(l_i^s = -1\right) \cdot \Pr\left(l_i^t = -1\right) \\ &= \left(\tfrac{1}{2}\right)^2 = \tfrac{1}{4}. \end{aligned} \tag{4}$$

Therefore, the probability that $C_i^{(2)} = 1$ is

$$\begin{aligned} \Pr\left(C_i^{(2)} = 1\right) &= 1 - \Pr\left(C_i^{(2)} = -1\right) \\ &= 1 - \tfrac{1}{4} = \tfrac{3}{4}. \end{aligned} \tag{5}$$

Since $P_{\text{QMAX2,3SAT}}$ contains m conjunctive clauses of the form $C_i^{(2)}$, the probability that all of them are satisfied is

$$\Pr\left(\wedge_{i=1}^{m} C_i^{(2)} = 1\right) = \left(\tfrac{3}{4}\right)^m . \tag{6}$$

By the same reasoning, for a third-order clause:

$$\begin{aligned}\Pr\left(\wedge_{i=1}^{n} C_i^{(3)} = 1\right) &= \left(1 - \Pr\left(C_i^{(3)} = -1\right)\right)^n \\ &= (1-(\frac{1}{2})^3)^n \\ &= \left(1 - \tfrac{1}{8}\right)^n = \left(\frac{7}{8}\right)^n ,\end{aligned} \tag{7}$$

Similarly, for a fourth-order clause:

$$\begin{aligned}\Pr\left(\wedge_{i=1}^{u} C_i^{(4)} = 1\right) &= \left(1 - \Pr\left(C_i^{(4)} = -1\right)\right)^u \\ &= (1-(\frac{1}{2})^4)^u \\ &= \left(1 - \tfrac{1}{16}\right)^u = \left(\frac{15}{16}\right)^u .\end{aligned} \tag{8}$$

Each $\left(C_{i1}^{(3)} \wedge C_{i2}^{(3)}\right)$ in $P_{\text{QMAX2,3SAT}}$ is equivalent to second-order clause. Therefore, according to (5),

$$\begin{aligned}\Pr\left(\wedge_{i=1}^{w}\left(C_{i1}^{(3)} \wedge C_{i2}^{(3)}\right) = 1\right) &= \Pr\left(\wedge_{i=1}^{w} C_i^{(2)} = 1\right) \\ &= \left(\frac{3}{4}\right)^w .\end{aligned} \tag{9}$$

According to Table 2, the probability that the induced neural configuration achieves the *maximum clause satisfaction* of the P_{QMAX2SAT} is

$$\Pr\left(\theta(P_{\text{QMAX2SAT}}) = \theta_{\max}\right) = \left(1 - \left(\frac{2}{8}\right)\right)^v = \left(\frac{3}{4}\right)^v . \tag{10}$$

Combining equations (6), (7), (8), (9) and (10), we can obtain

$$\begin{aligned}&\Pr\left(\theta(P_{\text{QMAX2,3SAT}}) = \theta_{\max}\right) \\ &= \Pr\left(\wedge_{i=1}^{m} C_i^{(2)} = 1\right) \cdot \Pr\left(\wedge_{i=1}^{m} C_i^{(3)} = 1\right) \cdot \Pr\left(\wedge_{i=1}^{m} C_i^{(4)} = 1\right) \cdot \Pr\left(\wedge_{i=1}^{w}\left(C_{i1}^{(3)} \wedge C_{i2}^{(3)}\right) = 1\right) \\ &\quad \cdot \Pr\left(\theta(P_{\text{QMAX2SAT}}) = \theta_{\max}\right) \\ &= \left(\frac{3}{4}\right)^m \cdot \left(\frac{7}{8}\right)^n \cdot \left(\frac{15}{16}\right)^u \cdot \left(\frac{3}{4}\right)^w \cdot \left(\frac{3}{4}\right)^v \\ &= \left(\frac{3}{4}\right)^{v+w+m} \left(\frac{7}{8}\right)^n \left(\frac{15}{16}\right)^u .\end{aligned} \tag{11}$$

This completes the proof. □

2.3. Proof of Theorem 3

Theorem 3. *Under random initialization of neuron states, the probability that the cost function of $P_{QMAX2,3SAT}$ attains its* minimum value *(*min*) is given by*

$$\Pr\Big(E_{P_{QMAX2,3SAT}} = \min\Big) = \left(\frac{3}{4}\right)^{v+w+m} \left(\frac{7}{8}\right)^{n} \left(\frac{15}{16}\right)^{u}, \tag{12}$$

where v, w, m, n, and u denote the numbers of the corresponding logical components integrated within $P_{QMAX2,3SAT}$. Moreover, the minimum value of the non-constant term in the cost function $E_{P_{QMAX2,3SAT}}$ is expressed as

$$E^{\min}_{P_{QMAX2,3SAT}} = -\frac{4v + 4w + 4m + 2n + u}{16}. \tag{13}$$

Proof. According to cost fuction $E_{P_{\mathrm{QMAX2,3SAT}}}$, for a second-order clause $C_i^{(2)} = l_i^s \vee l_i^t$ we have $\neg C_i^{(2)} = \neg l_i^s \wedge \neg l_i^t$ and

$$\begin{aligned}\prod_{j=1}^{2} O_{ij} &= \frac{1}{2}\left(1 - S_{l_i^s}\right) \cdot \frac{1}{2}\left(1 - S_{l_i^t}\right) \\ &= \frac{1}{4}\left(1 - S_{l_i^s} - S_{l_i^t} + S_{l_i^s} S_{l_i^t}\right) \\ &= \frac{1}{4} + \frac{1}{4}\left(-S_{l_i^s} - S_{l_i^t} + S_{l_i^s} S_{l_i^t}\right).\end{aligned} \tag{14}$$

Note that each factor $\frac{1}{2}(1 - S_l)$ equals 1 when the literal l is *false* and 0 when l is *true*, hence $\prod_{j=1}^{2} O_{ij} \in \{0, 1\}$. The event $\prod_{j=1}^{2} O_{ij} = 1$ occurs precisely when both $S_{l_i^s} = -1$ and $S_{l_i^t} = -1$, so

$$\begin{aligned}\Pr\left(\prod_{j=1}^{2} O_{ij} = 0\right) &= 1 - \Pr\left(\prod_{j=1}^{2} O_{ij} = 1\right) \\ &= 1 - \Pr\left(S_{l_i^s} = -1\right) \cdot \Pr\left(S_{l_i^t} = -1\right) \\ &= 1 - \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4}.\end{aligned} \tag{15}$$

When $\prod_{j=1}^{2} O_{ij} = 0$, the non-constant part of (14) attains its minimum value $-\frac{1}{4}$, so

$$\left(\prod_{j=1}^{2} O_{ij}\right)^{\min} = -\frac{1}{4} \tag{16}$$

Since $P_{\mathrm{QMAX2,3SAT}}$ contains m conjunctive clauses of the form $C_i^{(2)}$, the probability that all these terms contribute zero (i.e. all corresponding products equal 0) is

$$\Pr\left(\sum_{i=1}^{m} \prod_{j=1}^{2} O_{ij} = 0\right) = \left(\tfrac{3}{4}\right)^{m}, \tag{17}$$

and the minimum possible contribution from these terms is

$$\left(\sum_{i=1}^{m} \prod_{j=1}^{2} O_{ij}\right)^{\min} = -\frac{m}{4}. \tag{18}$$

By the same reasoning, for third-order clauses $C_i^{(3)}$ and fourth-order clauses $C_i^{(4)}$ one obtains

$$\begin{cases} \Pr\left(\sum_{i=1}^{n}\prod_{j=1}^{3} O_{ij} = 0\right) = \left(\frac{7}{8}\right)^n, \\ \left(\sum_{i=1}^{n}\prod_{j=1}^{3} O_{ij}\right)^{\min} = -\frac{n}{8}, \end{cases} \tag{19}$$

and

$$\begin{cases} \Pr\left(\sum_{i=1}^{u}\prod_{j=1}^{4} O_{ij} = 0\right) = \left(\frac{15}{16}\right)^u, \\ \left(\sum_{i=1}^{u}\prod_{j=1}^{4} O_{ij}\right)^{\min} = -\frac{u}{16}. \end{cases} \tag{20}$$

Now consider a pair of redundant third-order clauses of the form $C_{i1}^{(3)} \wedge C_{i2}^{(3)} = (l_i^a \vee l_i^b \vee l_i^c) \wedge (l_i^a \vee l_i^b \vee \neg l_i^c)$. Its negation is $\neg\left(C_{i1}^{(3)} \wedge C_{i2}^{(3)}\right) = (\neg l_i^a \wedge \neg l_i^b \wedge \neg l_i^c) \vee (\neg l_i^a \wedge \neg l_i^b \wedge l_i^c)$, hence the corresponding sum of products reduces to

$$\begin{aligned} \sum_{i=1}^{2}\prod_{j=1}^{3} O_{ij} =& \frac{1}{2}\left(1 - S_{l_i^a}\right)\cdot\frac{1}{2}\left(1 - S_{l_i^b}\right)\cdot\frac{1}{2}\left(1 - S_{l_i^c}\right) + \frac{1}{2}\left(1 - S_{l_i^a}\right)\cdot\frac{1}{2}\left(1 - S_{l_i^b}\right)\cdot\frac{1}{2}\left(1 + S_{l_i^c}\right) \\ =& \frac{1}{2}\left(1 - S_{l_i^a}\right)\cdot\frac{1}{2}\left(1 - S_{l_i^b}\right) \\ =& \frac{1}{4}\left(1 - S_{l_i^a} - S_{l_i^b} + S_{l_i^a}S_{l_i^b}\right). \end{aligned} \tag{21}$$

Thus each such redundant pair behaves like a single second-order clause with the same probability and minimum contribution; consequently, for w such pairs we have

$$\begin{cases} \Pr\left(\sum_{i=1}^{2w}\prod_{j=1}^{3} O_{ij} = 0\right) = \left(\frac{3}{4}\right)^w, \\ \left(\sum_{i=1}^{2w}\prod_{j=1}^{3} O_{ij}\right)^{\min} = -\frac{w}{4}. \end{cases} \tag{22}$$

Consider the example of $P_{\text{QMAX2SAT}} = (\neg l_i^p \vee \neg l_i^q) \wedge (\neg l_i^p \vee l_i^q) \wedge (l_i^p \vee \neg l_i^q) \wedge (l_i^p \vee \neg l_i^r) \wedge (l_i^q \vee l_i^r)$ in Table 2, its negation is $(l_i^p \wedge l_i^q) \vee (l_i^p \wedge \neg l_i^q) \vee (\neg l_i^p \wedge l_i^q) \vee (\neg l_i^p \wedge l_i^r) \vee (\neg l_i^q \wedge \neg l_i^r)$. Hence, the corresponding sum of products is

$$\begin{aligned} \sum_{i=1}^{5}\prod_{j=1}^{2} O_{ij} =& \frac{1}{2}\left(1 + S_{l_i^p}\right)\cdot\frac{1}{2}\left(1 + S_{l_i^q}\right) + \frac{1}{2}\left(1 + S_{l_i^p}\right)\cdot\frac{1}{2}\left(1 - S_{l_i^q}\right) + \frac{1}{2}\left(1 - S_{l_i^p}\right)\cdot\frac{1}{2}\left(1 + S_{l_i^q}\right) \\ &+ \frac{1}{2}\left(1 - S_{l_i^p}\right)\cdot\frac{1}{2}\left(1 + S_{l_i^r}\right) + \frac{1}{2}\left(1 - S_{l_i^q}\right)\cdot\frac{1}{2}\left(1 - S_{l_i^r}\right) \\ =& \frac{1}{4}\left(5 - S_{l_i^p}S_{l_i^q} - S_{l_i^p}S_{l_i^r} + S_{l_i^q}S_{l_i^r}\right). \end{aligned} \tag{23}$$

According to Table 2,

$$\begin{cases} \Pr\left(\sum_{i=5}^{v}\prod_{j=1}^{2} O_{ij} = \min = 1\right) = \left(\frac{3}{4}\right)^{v}, \\ \left(\sum_{i=1}^{5v}\prod_{j=1}^{2} O_{ij}\right)^{\min} = -\frac{v}{4}. \end{cases} \tag{24}$$

Combining all contributions (17), (18), (19), (20), (22) and (24), the probability that the cost function of $P_{\text{QMAX2,3SAT}}$ attains its minimum value (min) is

$$\begin{aligned} &\Pr\left(E_{P_{\text{QMAX2,3SAT}}} = \min\right) \\ &= \left(\frac{3}{4}\right)^{m} \cdot \left(\frac{7}{8}\right)^{n} \cdot \left(\frac{15}{16}\right)^{u} \cdot \left(\frac{3}{4}\right)^{w} \cdot \left(\frac{3}{4}\right)^{v} \\ &= \left(\frac{3}{4}\right)^{v+w+m} \left(\frac{7}{8}\right)^{n} \left(\frac{15}{16}\right)^{u} \end{aligned} \tag{25}$$

and the minimum the non-constant part is

$$\begin{aligned} E^{\min}_{P_{\text{ZRAN2,3SAT}}} &= -\frac{m}{4} - \frac{n}{8} - \frac{u}{16} - \frac{w}{4} - \frac{v}{4} \\ &= -\frac{4v + 4w + 4m + 2n + u}{16}. \end{aligned} \tag{26}$$

This completes the proof. □

2.4. Proof of Theorem 4

Theorem 4. *Under the asynchronous update governed by the local field and the activation function, the energy function $H_{P_{QMAX2,3SAT}}$ monotonically decreases and converges to a stable minimum, corresponding to an optimal information fusion state.*

Proof is given in Supplementary Material.

Proof. The energy of $H_{P_{\text{ZRAN2,3SAT}}}(t)$ associated with neuron S_i at time instants t and $t+1$ can be expressed as:

$$\begin{aligned} &H_{P_{\text{ZRAN2,3SAT}}}(t) \\ &= -\frac{1}{4}\sum_{i=1}^{n}\sum_{j=1}^{n}\sum_{k=1}^{n}\sum_{l=1}^{n} W^{(4)}_{ijkl} S_i(t) S_j S_k S_l - \frac{1}{3}\sum_{i=1}^{n}\sum_{j=1}^{n}\sum_{k=1}^{n} W^{(3)}_{ijk} S_i(t) S_j S_k \\ &- \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} W^{(2)}_{ij} S_i(t) S_j - \sum_{i=1}^{n} W^{(1)}_{i} S_i(t), \end{aligned} \tag{27}$$

and

$$\begin{aligned} &H_{P_{\text{ZRAN2,3SAT}}}(t+1) \\ &= -\frac{1}{4}\sum_{i=1}^{n}\sum_{j=1}^{n}\sum_{k=1}^{n}\sum_{l=1}^{n} W^{(4)}_{ijkl} S_i(t+1) S_j S_k S_l - \frac{1}{3}\sum_{i=1}^{n}\sum_{j=1}^{n}\sum_{k=1}^{n} W^{(3)}_{ijk} S_i(t+1) S_j S_k \\ &- \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} W^{(2)}_{ij} S_i(t+1) S_j - \sum_{i=1}^{n} W^{(1)}_{i} S_i(t+1). \end{aligned} \tag{28}$$

Similarly, the energy of $H_{P_{\text{QMAX2SAT}}}(t)$ associated with neuron i at time instants t and $t+1$ also can be expressed as:

$$H_{P_{\text{QMAX2SAT}}}(t) = -\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} W_{ij}^{(2)} S_i(t) S_j, \tag{29}$$

and

$$H_{P_{\text{QMAX2SAT}}}(t+1) = -\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} W_{ij}^{(2)} S_i(t+1) S_j. \tag{30}$$

Hence, the variation in energy between time t to $t+1$ can be written as:

$$\begin{aligned}
&\Delta H_{P_{\text{ZRAN2,3SAT}}} \\
&= H_{P_{\text{ZRAN2,3SAT}}}(t+1) - H_{P_{\text{ZRAN2,3SAT}}}(t) \\
&= -\left(S_i(t+1) - S_i(t)\right)\left(\sum_{j=1}^{n}\sum_{k=1}^{n}\sum_{l=1}^{n} W_{ijkl}^{(4)} S_j S_k S_l + \sum_{j=1}^{n}\sum_{k=1}^{n} W_{ijk}^{(3)} S_j S_k + \sum_{j=1}^{n} W_{ij}^{(2)} S_j + W_i^{(1)}\right) \\
&= -\left(S_i(t+1) - S_i(t)\right) h_i^{(Z)}(t),
\end{aligned} \tag{31}$$

and

$$\begin{aligned}
\Delta H_{P_{\text{QMAX2SAT}}} &= H_{P_{\text{QMAX2SAT}}}(t+1) - H_{P_{\text{QMAX2SAT}}}(t) \\
&= -\left(S_i(t+1) - S_i(t)\right)\left(\sum_{j=1}^{n} W_{ij}^{(2)} S_j\right) \\
&= -\left(S_i(t+1) - S_i(t)\right) h_i^{(M)}(t),
\end{aligned} \tag{32}$$

From the activation function, (31) and (32), it follows that

$$\Delta H_{P_{\text{QMAX2,3SAT}}} = -\left(S_i(t+1) - S_i(t)\right) h_i(t). \tag{33}$$

Since neuron i is arbitrary and all neurons update following the same rule, the total energy change is less than or equal to zero, expressed as:

$$H_{P_{\text{QMAX2,3SAT}}}(t+1) \le H_{P_{\text{QMAX2,3SAT}}}(t). \tag{34}$$

Since $H_{P_{\text{QMAX2,3SAT}}}$ is a bounded function, according to the fact that a monotonically decreasing bounded function must have a lower limit, we can obtain $H_{P_{\text{QMAX2,3SAT}}}$ will converge to a minimum value.

This completes the proof. □